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B.Tech. Degree VIII Semester Examination April 2016

CS 1801 ADVANCED ARCHITECTURE AND PARALLEL PROCESSING (2012 scheme)

Time : 3 Hours

Maximum Marks : 100

PART A (Answer *ALL* questions)

(8 × 5 = 40)

- I. (a) Explain Flynn's classification of computer architecture.
- (b) Explain the different types of data dependencies.
- (c) Difference the CISC and RISC processors.
- (d) Explain backplane bus specification.
- (e) Explain store load forwarding and load-load forwarding.
- (f) Explain wormhole routing.
- (g) What is Tomasulo's algorithm?
- (h) Explain multiprocessing, multitasking, multiprogramming and multithreading.



PART B

(4 × 15 = 60)

- II. (a) Explain the different program flow mechanisms and compare them. (10)
- (b) Explain UMA models. (5)

OR

- III. Explain any four static connection networks and any three dynamic connection networks. (15)

- IV. Explain super scalar and vector processors. (15)

OR

- V. (a) Describe cache memory organization using various mappings. (10)
- (b) Explain Lamport's definition for sequential consistency. (5)

- VI. (a) Explain the working of an asynchronous and synchronous pipeline processor. (5)
- (b) Define speedup, efficiency and thrashput of a linear pipeline of 'k' stages. (5)
- (c) Explain the cache coherence problem. (5)

OR

- VII. (a) Explain the following architectures (7)
 - (i) Crossbar switch
 - (ii) Multiport memory.
- (b) Explain the SIMD computer organizations. (8)

- VIII. (a) Explain any three parallel programming models. (7)
- (b) Explain local code optimization methods. (8)

OR

- IX. (a) Explain the models of a typical processor (10)
- (b) Explain what you mean by branch prediction. (5)

